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A PROJECT PHASE II REPORT

On

Design Simulation and Testing of 3.5 GHz Band-Pass FSS for 5G Application

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ABSTRACT

Frequency Selective Surfaces (FSS) are periodic structures that act as spatial filters for electromagnetic waves, enabling selective transmission or reflection of specific frequency bands. In modern wireless communication systems, especially in 5G Sub-6 GHz applications, there is a growing need to efficiently transmit desired signals while suppressing unwanted interference. This project presents the design, simulation, and experimental validation of a 3.5 GHz band-pass Frequency Selective Surface suitable for 5G applications.

The proposed FSS structure is designed using a periodic unit cell configuration and analyzed using electromagnetic simulation software. The unit cell is optimized to achieve resonance at 3.5 GHz, which lies within the widely used 5G frequency band. The design is extended to an array configuration to replicate practical implementation conditions. Parametric analysis is carried out to ensure optimal performance in terms of reflection and transmission characteristics.

The simulated results demonstrate that the proposed FSS exhibits a reflection coefficient (S11) of approximately –21 dB and a transmission coefficient (S21) close to 0 dB at 3.5 GHz, indicating excellent impedance matching and high transmission efficiency. Additionally, angular stability analysis is performed by varying the incident angles (phi and theta), confirming consistent performance across a wide range of angles. This ensures the reliability of the design in real-world scenarios where signals arrive from different directions.

To validate the simulation results, the FSS structure is fabricated using standard PCB techniques. A practical measurement setup involving transmitting and receiving antennas along with a network analyzer is considered to evaluate the transmission characteristics of the fabricated prototype. The experimental approach verifies that the structure effectively allows the desired frequency while attenuating others, confirming its band-pass behavior.

The proposed FSS can be effectively used in 5G communication systems, radomes, and electromagnetic interference suppression applications, where selective frequency transmission is essential. The combination of compact design, good transmission characteristics, and angular stability makes the proposed structure suitable for practical wireless applications.

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FSS	-	Frequency Selective Surface
HFSS	-	High Frequency Structure Simulator

CHAPTER 1

INTRODUCTION

1.1 SCOPE AND OBJECTIVE

The scope of this project focuses on the design, simulation, fabrication, and validation of a 3.5 GHz band-pass Frequency Selective Surface (FSS) for modern wireless communication applications, particularly in dense urban environments. The project aims to develop a compact and efficient unit cell structure and extend it into an array configuration to achieve practical implementation. The primary objective is to ensure selective transmission at the desired frequency while suppressing unwanted signals, thereby improving signal quality and reducing interference. The design is optimized to achieve good impedance matching with a reflection coefficient around –21 dB and high transmission efficiency. Additionally, the project emphasizes angular stability by analyzing the performance over a wide range of incident angles (phi and theta up to 360°), ensuring reliable operation in real-world conditions where signals arrive from multiple directions. The fabricated prototype is experimentally validated using antennas and a network analyzer to confirm its band-pass behavior. Overall, the project contributes to applications such as 5G communication systems, smart buildings, and electromagnetic interference control in urban areas.

1.2 MOTIVATION:

The exponential growth in wireless communication technologies, particularly the deployment of 5G systems in the Sub-6 GHz spectrum, has led to increased challenges in managing electromagnetic interference and ensuring efficient spectrum utilization. The 3.5 GHz frequency band is widely adopted for 5G applications due to its balance between coverage and data rate; however, in densely populated urban environments, the presence of multiple transmitting sources and reflections from surrounding structures results in significant signal interference and degradation.

Conventional filtering techniques are often limited to circuit-level implementations and may not effectively control electromagnetic waves in free space. This limitation has driven the need for spatial filtering solutions that can selectively control the propagation of electromagnetic waves. Frequency Selective Surfaces (FSS) provide a promising approach by acting as periodic structures capable of transmitting or reflecting specific frequency bands while suppressing others.

Furthermore, real-world communication scenarios involve signals arriving from multiple directions due to multipath propagation, making angular stability an essential requirement for practical designs. This motivates the development of an FSS structure that not only operates efficiently at the desired frequency of 3.5 GHz but also maintains consistent performance across varying angles of incidence.

Therefore, this project is motivated by the need to design a compact, efficient, and angularly stable Frequency Selective Surface that can improve signal selectivity, reduce interference, and support reliable communication in modern 5G and urban wireless environments.

1.3 PROBLEM IDENTIFICATION

The rapid expansion of wireless communication systems, particularly 5G networks operating in the 3.5 GHz band, has led to increased challenges in maintaining signal integrity and efficient spectrum utilization. In modern communication environments, especially in dense urban areas, multiple wireless devices operate simultaneously within overlapping frequency bands. This results in significant electromagnetic interference, signal distortion, and degradation of communication quality.

Additionally, electromagnetic waves in real-world scenarios do not propagate in a single direction. Due to reflections, scattering, and diffraction caused by buildings and other obstacles, signals arrive at receiving systems from multiple angles. Conventional filtering techniques, which are typically implemented at the circuit level, are insufficient to control such spatial and directional signal variations effectively.

Another critical issue is the lack of compact and efficient structures that can selectively allow desired frequencies while suppressing unwanted signals in free-space environments. Existing solutions often fail to provide consistent performance over a wide range of incident angles, limiting their practical applicability.

Therefore, there is a need for a spatial filtering structure that can operate at the desired 3.5 GHz frequency, provide high transmission efficiency, suppress unwanted frequencies, and maintain stable performance for signals arriving from different directions. Addressing these challenges forms the basis of this project, which focuses on the design and implementation of a Frequency Selective Surface (FSS) to overcome the limitations of conventional approaches.

1.4 DOMAIN INTRODUCTION:

This project falls under the domain of Electromagnetics and Wireless Communication, which deal with the analysis, design, and application of electromagnetic wave propagation in various communication systems. With the rapid advancement of wireless technologies, particularly 5G communication systems, there is an increasing demand for efficient techniques to control and manage electromagnetic waves in free space.

One of the key areas within this domain is the study of Frequency Selective Surfaces (FSS), which are periodic structures capable of filtering electromagnetic waves based on frequency. These surfaces are widely used in RF and microwave engineering applications such as antennas, radomes, and electromagnetic shielding systems. FSS structures function similarly to electrical filters but operate in the spatial domain, allowing specific frequency bands to pass while reflecting or attenuating others.

The growing deployment of 5G networks, especially in the Sub-6 GHz frequency range around 3.5 GHz, has further emphasized the importance of such technologies. In real-world environments, particularly in urban areas, electromagnetic waves are subject to reflection, scattering, and multipath propagation, leading to interference and signal degradation. This necessitates the development of advanced filtering techniques that can operate efficiently in free-space conditions and maintain stable performance for signals arriving from different directions.

Therefore, this project focuses on the design and analysis of a Frequency Selective Surface operating at 3.5 GHz, contributing to the broader domain of RF and microwave engineering by addressing challenges related to signal selectivity, interference reduction, and angular stability in modern wireless communication systems.

1.5 FSS

A Frequency Selective Surface (FSS) is a two-dimensional periodic structure composed of conductive elements or apertures arranged on a dielectric substrate, designed to control the propagation of electromagnetic waves based on frequency. It functions as a spatial filter, allowing specific frequency bands to be transmitted while reflecting or attenuating others, similar to conventional electronic filters but operating in free space. The fundamental building block of an FSS is the unit cell, which is repeated in an array configuration to achieve the desired electromagnetic response. The performance of an FSS depends on various parameters such as the shape, size, orientation, and periodicity of the unit cell, as well as the properties of the substrate material.

FSS structures can be classified into band-pass or band-stop types depending on whether they allow or block a particular frequency range. These surfaces are widely utilized in applications such as antenna radomes, electromagnetic interference (EMI) shielding, satellite communication systems, and modern wireless networks. In recent years, with the advancement of 5G communication systems operating in the Sub-6 GHz band, particularly around 3.5 GHz, FSS has gained significant importance for improving signal selectivity and reducing interference.

In practical environments, especially in urban areas, electromagnetic waves experience reflection, diffraction, and scattering due to obstacles such as buildings and other structures. As a result, signals arrive from multiple directions, making angular stability a critical factor in FSS design. A well-designed FSS should maintain consistent performance over a wide range of incident angles to ensure reliable operation in real-world conditions. Due to its compact structure, passive nature, and ability to efficiently control electromagnetic wave propagation, FSS has become an essential component in modern RF and microwave engineering applications.

Basic Principle of Operation:

When a plane electromagnetic wave impinges on an FSS, part of the wave is reflected and part is transmitted. The ratio of reflection to transmission depends on the resonance characteristics of the structure. The resonant frequency is mainly determined by the unit cell dimensions, shape, periodicity, and substrate parameters**.**

At resonance:

If the structure supports strong surface currents (as in metallic patches), the surface behaves as a reflector (band-stop).If it supports strong electric field coupling through apertures (slots), it behaves as a transmitter (band-pass).

However, practical realizations often show deviations from these theoretical expectations due to size of structure, dielectric losses, mutual coupling, finite substrate thickness, and boundary conditions. Thus, the actual electromagnetic behavior (band-pass or band-stop) depends not only on the structure type but also on the complete configuration and simulation setup or results.

Band – Pass FSS

A Band-Pass Frequency Selective Surface (FSS) is a periodic structure designed to allow electromagnetic waves of a specific desired frequency to pass through while attenuating or blocking other unwanted frequencies. It functions as a spatial filter, enabling selective transmission in free space similar to a band-pass filter in circuits. This type of FSS is widely used in applications where efficient transmission of a particular frequency band is required, such as 5G communication systems, radomes, and signal filtering. In this project, the designed FSS operates at 3.5 GHz and exhibits band-pass characteristics by allowing the desired frequency to transmit with high efficiency while suppressing other frequencies.

Band – Stop FSS

A Band-Stop Frequency Selective Surface (FSS) is designed to block or reject a specific frequency band while allowing other frequencies to pass through. It reflects the targeted frequency back instead of transmitting it, thereby preventing its propagation. This type of FSS is useful in applications where certain frequencies need to be eliminated to avoid interference or detection. Band-stop FSS structures are commonly used in electromagnetic interference (EMI) shielding, radar systems, and interference suppression applications. By selectively rejecting unwanted frequencies, band-stop FSS improves overall system performance and signal quality.

Types of Frequency Selective Surfaces

FSSs are generally classified into two major categories based on their physical construction and electromagnetic behavior:

1. Patch-Type FSS

Composed of periodic metallic patches printed on a dielectric substrate.Acts as a band-stop filter, reflecting EM waves at specific resonant frequencies.The resonance occurs due to the induced surface currents on the metallic patches.Common patch geometries include square, circular, cross, Jerusalem cross, ring, and dipole shapes.Widely used in radome covers, frequency-rejecting screens, and reflectarrays.

2. Slot-Type (Aperture-Type) FSS

Formed by etching slots or apertures in a continuous metallic layer.
Typically behaves as a band-pass surface, allowing specific frequencies to pass while blocking others.Resonance arises from the coupling of electric fields through the slots.

Slot shapes can vary from rectangular and circular to complex geometries like cross-slots or star-slots.Commonly used in frequency-selective windows, absorbers, and polarization control structures.

Although these two types are theoretically complementary (as explained by Babinet's principle), in practical designs, the behavior can interchange based on substrate properties, slot configuration, or design purpose. Therefore, a slot-based FSS can exhibit band-stop behavior if it is designed to produce high reflection rather than transmission.

Design Considerations of FSS

The performance of an FSS depends on several critical design parameters:

1. Geometry of the Unit Cell:

The shape of the metallic patch or slot determines the resonance frequency and polarization characteristics. Common geometries include crosses, loops, rings, and star shapes.
Complex geometries allow multi-band or polarization-insensitive designs.

2. Periodicity (P):

The distance between adjacent unit cells influences the coupling between them. If the periodicity is small compared to the wavelength, mutual coupling is strong, leading to broader bandwidths.Larger periodicity can improve frequency selectivity but may introduce higher-order resonances.

3. Substrate Material:

The dielectric constant (εr) and thickness of the substrate affect the resonant frequency and bandwidth.A higher dielectric constant reduces the resonant frequency.A thicker substrate can improve bandwidth but may increase losses.

4. Metallic Layer Properties:

The conductivity and thickness of the metallic layer influence losses and the sharpness of resonance.Highly conductive materials such as copper or aluminum are preferred for low-loss operation.

5. Incidence Angle:

The FSS performance must remain stable for various angles of incidence.

6. Boundary Conditions:

Periodic boundary conditions are applied during simulation to replicate an infinite array, ensuring accurate frequency response predictions.

Electromagnetic Behavior and Resonance

The resonance mechanism in an FSS arises from the interaction between the incident EM field and the periodic elements. At resonance, strong induced currents or fields cause maximum reflection or transmission depending on structure type.

For patch-type FSS, surface currents dominate, resulting in reflection resonance (band-stop behavior).For slot-type FSS, electric field coupling through apertures dominates, leading to transmission at resonance (band-pass behavior).

However, when the slot geometry is complex or when the FSS is backed by a substrate that alters field distribution, the same slot-based design can exhibit reflection-dominant characteristics.

This principle explains why a slot-based FSS may show band-stop behavior if the electromagnetic fields are confined and reflected instead of transmitted — a phenomenon that is often used in modern reflective applications like 5G reflectors and reconfigurable surfaces.

Simulation and Analysis Parameters

i. To design and analyze an FSS, full-wave electromagnetic simulation HFSS tool is used. The key parameters evaluated are:
ii. S_{11} (Reflection Coefficient): Indicates how much of the incident signal is reflected.
iii. S_{21} (Transmission Coefficient): Indicates how much of the signal passes through the surface.
iv. The resonant frequency corresponds to the point where reflection or transmission peaks occur. A strong dip in S_{11} or peak in S_{21} indicates resonance, and the corresponding frequency defines the operating band of the FSS.

Applications of FSS

Application in 5G Communication Systems

i. Operates in 3.5 GHz Sub-6 GHz band widely used in modern wireless networks
ii. Use in Urban Environments

Suitable for dense urban areas where:

i. Helps in filtering unwanted frequencies

Smart Building Integration

Can be integrated into:

i. Windows
ii. Walls

Controls electromagnetic wave propagation inside buildings

i. RF and Microwave Systems

CHAPTER 2

LITERATURE SURVEY

With the rapid deployment in 5G networks, enhancing the coverage in wireless network and the signal in the urban areas remains a significant challenge due to signal blockage by the tall building and other obstacles. A 3-bit IRS based on modified square loop Frequency Selective Surfaces (FSS) integrated with variable capacitor diode demonstrated promising phase control and performance at 3.5 GHz typically used for sub-6 GHz purposes. This work laid important groundwork for practical IRS deployment in the sub-6 GHz band [1-3]. Prior research has explored IRS designs tailored for both mm Wave and sub-6 GHz frequencies. Millimeter-wave IRS implementations often focus on beam-steerable MIMO antennas and multi-band reflect array metasurfaces to enhance 5G coverage [3-4]. However, sub-6 GHz IRS designs face unique challenges such as wider angular incidence variations and signal penetration losses that require specialized design approaches.

A RIS, also referred to as an IRS, which recently considered as an emerging innovative technology can dynamically control the wireless propagation to improve communication performance [4-6]. IRS are realized using passive metasurfaces made of tunable unit cells. These cells regulate the phase of incoming electromagnetic waves and amplitude of incoming electromagnetic waves. They direct signals toward intended receivers without needing power-hungry active components like power amplifiers [7&8]. This feature makes IRS an energy-efficient and cost-effective option for future wireless networks at Multiple frequency bands, spanning both the lower sub-6 GHz range and the higher mm Wave spectrum.

Further studies have mentioned the importance of realistic wireless channel modelling and analysis the performance for IRS-based communications in sub-6 GHz scenarios, highlighting the need for robust designs capable of maintaining stable electromagnetic response under varying incidence angles [7-9]. Real-world demonstrations of IRS at sub-6 GHz confirm that angular stability and resonance robustness are critical to ensuring reliable coverage enhancement in urban environments. Based on these findings, this paper presents the design and simulation of an FSS unit cell for IRS optimized for the 3.5 GHz band a key frequency for 5G communications. The proposed unit cell is engineered to maintain consistent resonance and reflection performance across a range of phi and theta incidence angles, addressing the practical deployment challenges associated with signal degradation in obstructed urban areas [9 & 10]. Recent works have also explored broader metasurface and FSS-based approaches that complement IRS designs. For example, highly compact dual-band FSS structures have been reported to reduce path loss and extend wireless coverage in advanced communication systems [11]. Surveys on FSS and meta surfaces emphasize their increasing role not only in 5G but also paving the way for 6G networks, highlighting emerging design directions and practical applications [12]. Additionally, reconfigurable absorber concepts have shown the capability of switching between reflection and transmission bands, demonstrating the adaptability of FSS in dynamic wireless environments [13]. Other studies have proposed frequency-selective surfaces for high-gain MIMO antennas in sub-6 GHz bands, illustrating their effectiveness in beamforming and coverage enhancement [14]. Furthermore, real-world IRS implementations in sub-6 GHz frequencies confirm that these designs are both feasible and effective in enhancing signal coverage, supporting their potential for deployment in dense urban areas [15].

Reference	**Frequency (GHz)**	**Array**	**Angular Stability**	**Substrate**
[1]	3.5	-	60°	FR4
[3]	30	32×32	45°	RO3003
[4]	24 & 38	32×32	60°	RO3003
[5]	15.48	32×32	85°	RO4003C
[11]	24 & 38	32×32	60°	RT5880
Proposed	3.5	2×2	360°	FR4

Table 1 Comparison Table

The proposed design operates at 3.5 GHz with enhanced angular stability from 0° to 360°, surpassing existing works limited to 60°-85° which is showed in Table 1.

CHAPTER – 3
SYSTEM DESIGN & METHODOLOGY

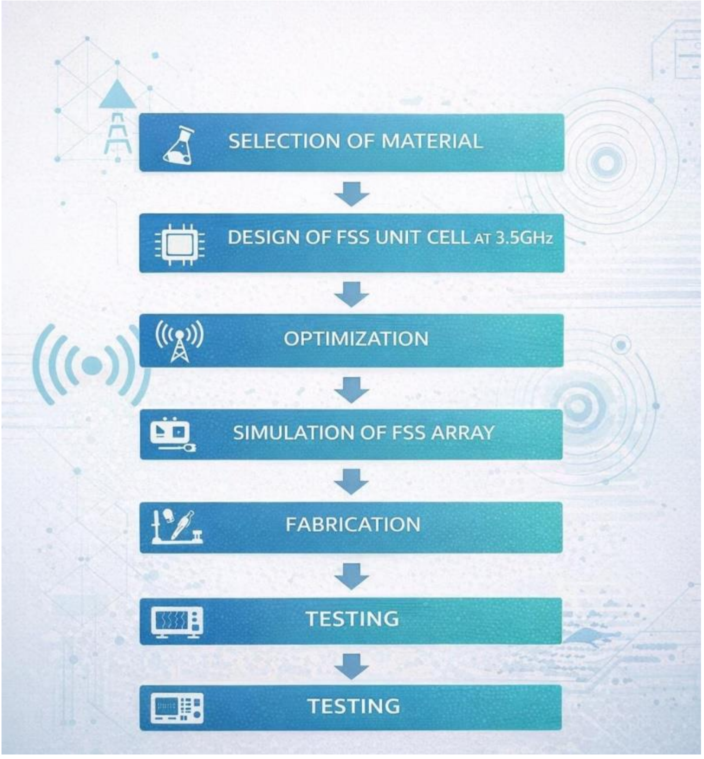


Fig. 1 Block Diagram

The overall system design of the proposed Frequency Selective Surface (FSS) follows a structured methodology, as illustrated in Fig. 1. The process begins with the selection of appropriate materials, followed by the design, simulation, optimization, fabrication, and testing of the FSS structure to achieve the desired performance at 3.5 GHz.

Initially, suitable substrate and conductive materials are selected based on their electrical properties such as dielectric constant, loss tangent, and conductivity. The choice of material plays a crucial role in determining the performance and efficiency of the FSS.

Next, the unit cell of the FSS is designed to resonate at the target frequency of 3.5 GHz. The geometry, dimensions, and periodicity of the unit cell are carefully chosen to obtain the desired band-pass characteristics. The designed unit cell is then analyzed using electromagnetic simulation software.

Following the initial design, optimization is carried out by varying different parameters such as dimensions and spacing of the unit cell to achieve improved transmission characteristics and better impedance matching. The goal is to obtain high transmission (S21 close to 0 dB) and low reflection (S11 close to –20 dB).

Once the optimized unit cell is obtained, it is extended into an array configuration to replicate practical implementation conditions. The FSS array is then simulated to analyze its overall performance and angular stability for different incident angles.

After successful simulation, the design is fabricated using standard PCB techniques. The fabricated prototype represents the practical realization of the proposed FSS structure.

Finally, testing is performed using a measurement setup that includes transmitting and receiving antennas along with a network analyzer. The experimental results are analyzed to verify the band-pass behavior and validate the simulated performance of the FSS.

3.1 UNIT CELL:

The proposed Frequency Selective Surface (FSS) unit cell is designed with a cross-shaped slot geometry that provides efficient frequency-selective behavior in the 3.5 GHz band. The structure consists of a metallic layer with the slot pattern etched on a dielectric substrate of length L, which acts as the supporting base material for the unit cell. This slot-based configuration allows accurate regulation of the reflection and transmission characteristics of incident electromagnetic waves.

The design is symmetrical along both the X and Y axes, ensuring uniform electromagnetic response in all directions. The cross-shaped slot extends outward from the center, with each arm tapering into sharp edges as shown in Fig 3. These pointed ends help in concentrating the surface current distribution, improving the resonance characteristics and frequency selectivity of the structure. The formation of unit cell is shown in fig 2.

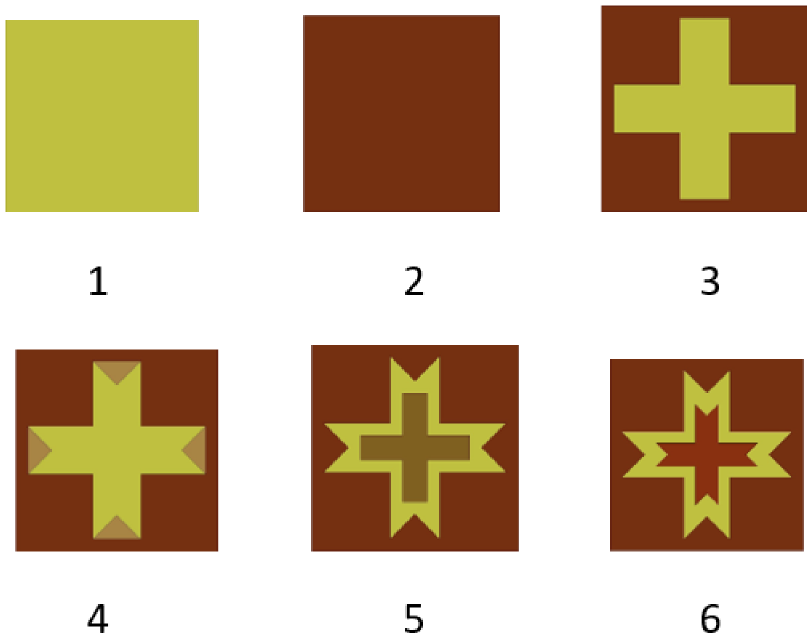


Fig.2 Formation of Unit Cell

The overall substrate length (L) defines the physical dimension of each unit cell and plays a key role in determining the resonant frequency. By optimizing parameters such as arm width, slot length, and spacing as shown in Table 2, these cells are arranged periodically in along the X- and Y-axes, the surface behaves as a band-pass FSS, effectively reflecting signals within the desired frequency range while minimizing transmission, the structure achieves stable electromagnetic performance at the desired operating frequency.

To ensure accurate and realistic electromagnetic simulation, suitable boundary conditions and excitation setups were applied.

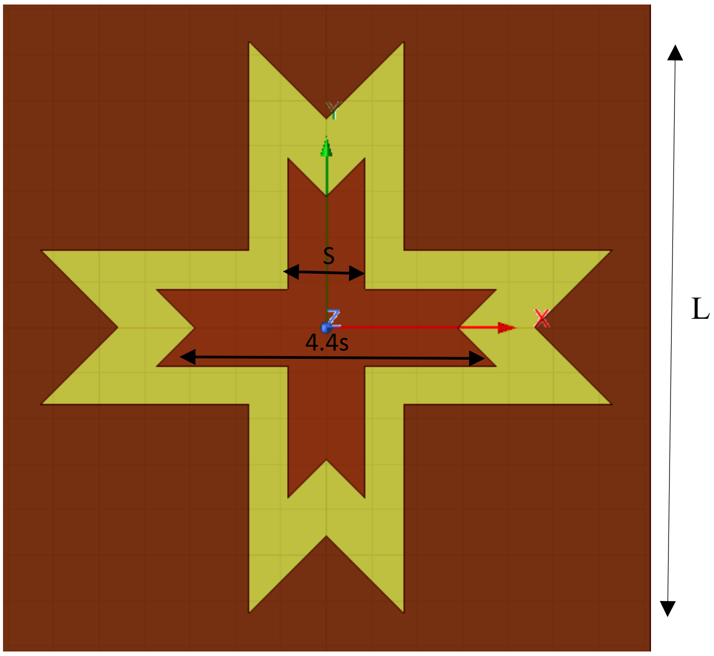


Fig.3 Unit Cell

Boundary Conditions

The boundary assignment in HFSS plays a vital role in representing the periodic nature of an FSS structure. Since the FSS is made up of a repeated pattern of identical unit cells, it is not necessary to simulate the entire array. Instead, periodic boundaries are used to model the infinite repetition of the unit cell.

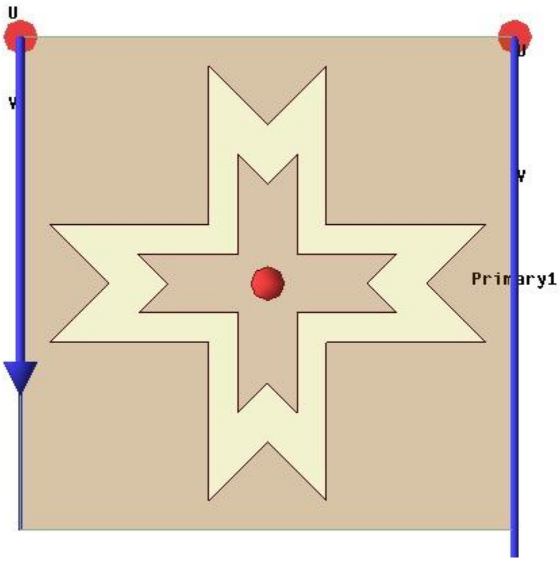


Fig.4 Primary 1

In this design, Master and Slave boundaries—also referred to as Primary and Secondary boundaries—were applied on the vertical sides of the unit cell as shown in fig 4 & fig 5.

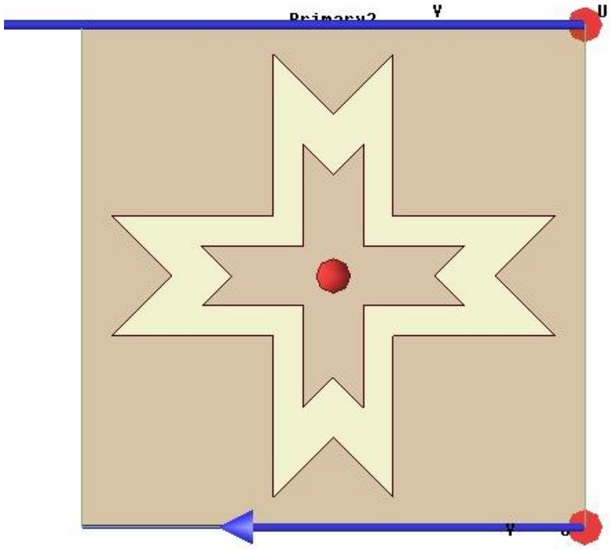


Fig.5 Primary 2

These boundaries ensure that the electromagnetic fields at one side of the unit cell are identical to those on the opposite side, maintaining periodicity in both the x and y directions. This setup effectively simulates an infinite array, reducing computational complexity while preserving the physical accuracy of the results.

By implementing these boundaries, the structure accurately represents the behavior of a large FSS surface under plane wave excitation, allowing the software to calculate reflection and transmission parameters without simulating the entire array physically.

Excitations

To excite the electromagnetic wave on the FSS surface, Floquet ports were assigned to the top and bottom surfaces of the unit cell as shown in fig 6 & 7. Floquet ports are specifically used in periodic structures because they can handle multiple diffraction modes of electromagnetic waves.

The top port acts as the incident wave source, while the bottom port captures the reflected and transmitted waves. The port configuration allows the calculation of S-parameters (S11 and S21), representing reflection and transmission coefficients, respectively.

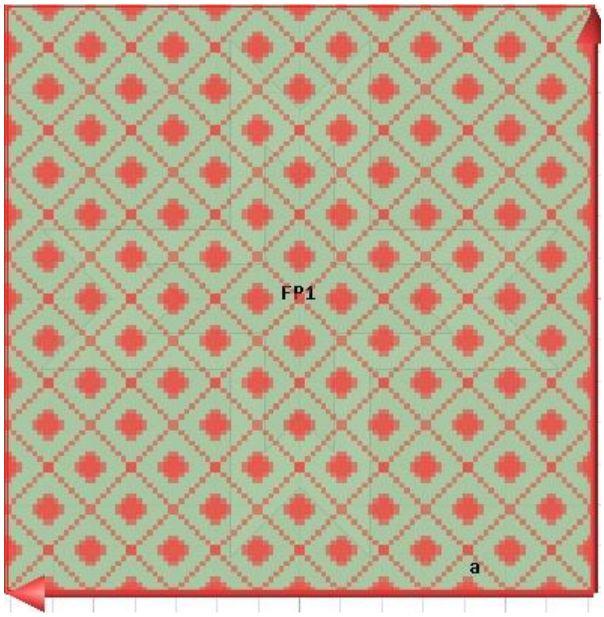


Fig.6 Floquet port 1

This combination of Master/Slave periodic boundaries and Floquet port excitations ensures a realistic electromagnetic environment for simulation. It accurately captures the periodic response, angular stability, and reflection characteristics of the designed FSS.

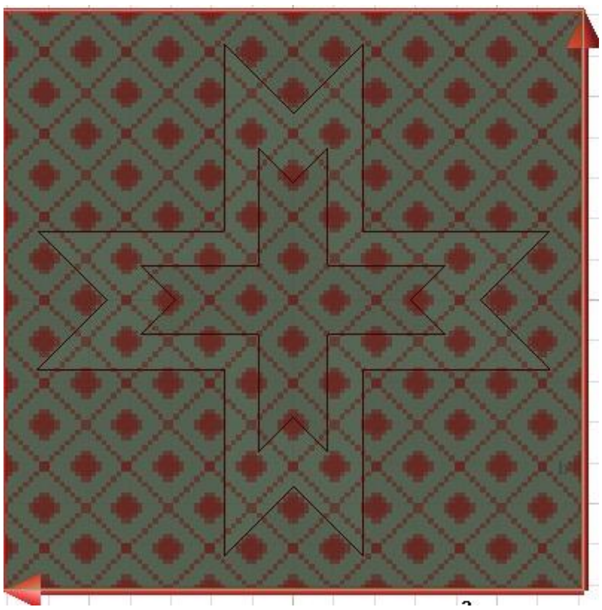


Fig.7 Floquet port 2

This optimized cross-slot structure, supported on a dielectric substrate of length L, is highly suitable for IRS applications. It provides compactness, stable frequency response, and strong reflection capability, making it a valuable design for 5G and advanced wireless communication systems.

Parameter	Value
Operating Frequency	3.5GHz
Substrate Material	FR - 4
Substrate Thickness(h)	1.6mm
Period (L)	28.54mm
Slot Width (S)	3.4mm
Array	2x2
Array Size	57.08mm

Table 2 FSS Parameter

3.2 ANGULAR STABILITY:

The angular stability of a Frequency Selective Surface (FSS) is a crucial factor that determines how consistently the structure performs when electromagnetic waves strike it from different angles. In real-world wireless communication environments, the incident wave rarely arrives perpendicularly; instead, it comes from various directions depending on where the transmitter and receiver are located. Therefore, it is essential that the FSS maintains its reflection characteristics across different angles of incidence to ensure stable performance.

In this project, the proposed cross-shaped unit cell was analyzed for angular stability by varying the incident angle (θ and φ) from 0° to 360°. The analysis was performed by observing only the s_{11} parameter, which indicates the reflection capability of the structure at different angles.

The simulation results showed that the s_{11} response remained stable over the entire range of angular variations, confirming that the unit cell maintains a consistent reflection behavior even when the angle of incidence changes. This demonstrates that the design possesses good angular stability, ensuring that its performance is not significantly affected by directional variations in the incoming signal.

The excellent angular stability of the proposed FSS can be attributed to its symmetrical cross-slot geometry, which provides uniform current distribution and balanced electromagnetic response along both the X and Y axes. Such stable reflection characteristics make the structure highly suitable for Intelligent Reflecting Surface (IRS) applications, where signals are reflected from multiple directions to improve coverage and efficiency in 5G communication systems.

3.3 ARRAY:

After analyzing the performance of the proposed unit cell, an array structure was developed to study the collective behavior of multiple cells and to enhance the overall reflection performance of the Frequency Selective Surface (FSS). The array helps to closely replicate real-world FSS operation, where a large number of periodic elements work together to reflect or control electromagnetic waves efficiently.

In this project, a 2 × 2 matrix of the optimized cross-shaped unit cell was constructed which is shown in fig 8. The periodic arrangement was achieved by repeating the single unit cell pattern along both the X and Y axes, maintaining uniform spacing between adjacent elements to preserve proper phase alignment and coupling. This configuration ensures that the reflected waveforms from each unit cell combine coherently, resulting in improved reflection efficiency and surface uniformity.

The array structure allows better study of mutual coupling effects between neighboring elements and helps verify that the designed geometry maintains consistent reflection properties when expanded into a larger surface. The symmetrical layout of the unit cells also ensures that the structure remains angularly stable and directionally consistent.

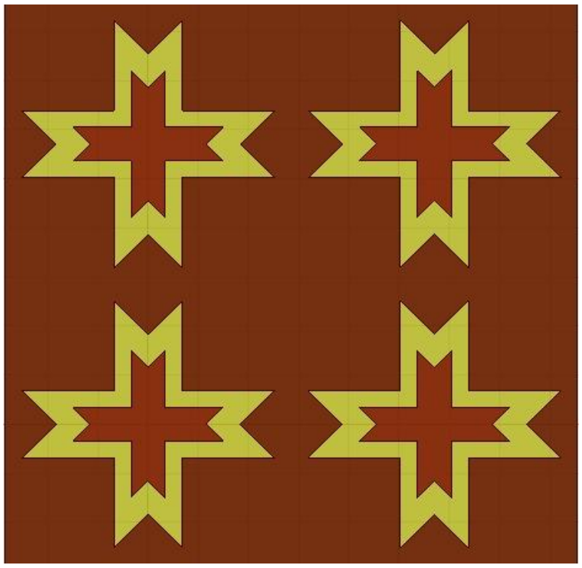


Fig.8 Array Structure

Such an array-based design approach is crucial for IRS applications, where multiple reflecting elements are used to dynamically control and redirect the electromagnetic waves toward desired directions. The developed 2 × 2 FSS array serves as a practical model for implementing larger reflective panels in 5G wireless communication systems, offering enhanced signal coverage, better beam control, and efficient frequency selectivity.

3.4 CURRENT ANALYSIS:

To better understand the electromagnetic behavior of the proposed Frequency Selective Surface (FSS), surface current analysis is performed at the resonant frequency of 3.5 GHz. The distribution of current on the metallic surface is studied using both vector surface current (vector Jsurf) and magnitude surface current (mag Jsurf) plots.

The vector Jsurf plot represents the direction of current flow on the surface of the unit cell as shown in fig 9. It helps in identifying how the current circulates through different parts of the structure. From the analysis, it is observed that the current is mainly concentrated along the edges and specific segments of the conducting pattern, indicating the active regions responsible for resonance.

The magnitude Jsurf plot shows the intensity of the surface current distribution as shown in fig 10. Higher current density regions correspond to stronger electromagnetic interaction and play a significant role in determining the resonant behavior of the FSS.

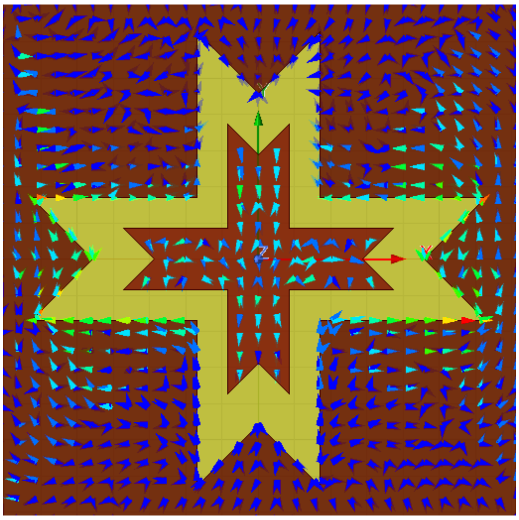


Fig. 9 Vector_Jsurf

At 3.5 GHz, the maximum current concentration is observed in critical portions of the structure, confirming that the designed geometry effectively supports resonance at the desired frequency.

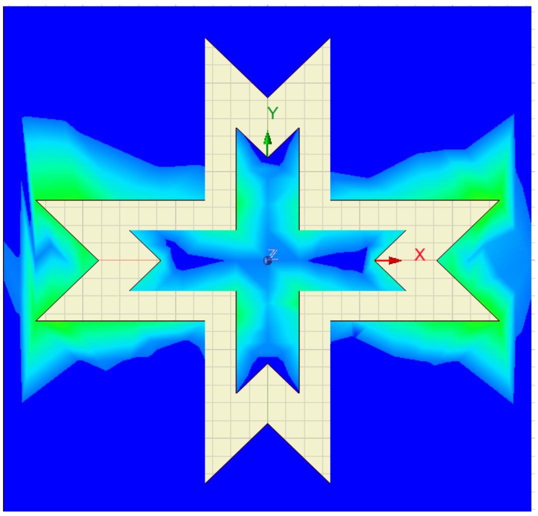


Fig. 10 Mag_Jsurf

This analysis provides clear insight into the working mechanism of the FSS and validates that the structure is properly designed to achieve band-pass characteristics. The concentration and flow of surface currents directly influence the transmission and reflection properties, thereby confirming the effectiveness of the proposed design.

CHAPTER 4

IMPLEMENTATION AND DEVELOPMENT

The implementation of the proposed Frequency Selective Surface (FSS) is carried out using electromagnetic simulation software, followed by fabrication and testing. The development process includes the design of the unit cell, application of boundary conditions, excitation setup, parametric analysis, array formation, angular analysis, and performance evaluation using S-parameters.

Initially, the FSS unit cell is designed to operate at a target frequency of 3.5 GHz. The structure consists of a periodic metallic pattern printed on a dielectric substrate. The geometry and dimensions of the unit cell are carefully selected to achieve the desired band-pass response. The design is modeled in the simulation environment with appropriate material properties assigned to both the substrate and conducting layer.

To replicate an infinite periodic structure, boundary conditions are applied. Master-slave (or periodic) boundary conditions are used along the lateral directions of the unit cell, ensuring that the structure behaves as an infinite array. Additionally, radiation boundaries are applied to allow electromagnetic waves to propagate without reflection from the simulation domain.

For excitation, wave ports (Floquet ports) are used to simulate plane wave incidence on the FSS structure. This setup enables accurate analysis of transmission and reflection characteristics. The excitation is applied normal to the surface initially, and later extended to different angles for angular stability analysis.

A frequency sweep is performed over a range covering the desired operating band (typically from 2.5 GHz to 4.5 GHz). This helps in identifying the resonant frequency and analyzing the S-parameters of the structure. The simulated results show that the structure resonates at approximately 3.5 GHz.

The unit cell is then extended into an array configuration to represent practical implementation. The array ensures that mutual coupling effects and realistic behavior of the FSS are taken into account.

Further analysis is carried out by varying the incident angles (phi and theta) to evaluate angular stability. The results indicate that the proposed FSS maintains consistent performance over a wide range of angles, confirming its suitability for real-world applications where signals arrive from multiple directions.

The performance of the FSS is evaluated using S-parameters. The reflection coefficient (S11) is observed to be approximately –21 dB at 3.5 GHz, indicating good impedance matching and minimal reflection. The transmission coefficient (S21) is close to 0 dB, confirming high transmission efficiency at the desired frequency.

The bandwidth of the proposed FSS is calculated from the –10 dB points and is found to be approximately 283 MHz, indicating a stable and efficient band-pass response.

Overall, the implementation and development of the proposed FSS demonstrate that the structure achieves the desired performance in terms of frequency selectivity, transmission efficiency, and angular stability, making it suitable for practical applications in 5G communication systems and urban environments.

CHAPTER 5

RESULTS & DISCUSSION

5.1 Unit Cell Result:

The simulated unit cell structure of the proposed Frequency Selective Surface (FSS) was analyzed to evaluate its reflection and transmission characteristics using HFSS software. The analysis was performed over the frequency range of 2.5 GHz to 4.5 GHz to ensure that the designed structure resonates precisely at the desired operating frequency for sub-6 GHz 5G applications.

The S-parameter response, particularly S_{11} (reflection coefficient) and S_{21} (transmission coefficient), was examined to assess the electromagnetic behavior of the unit cell. From the simulation results, the return loss (S_{11}) was found to be –21 dB at the resonant frequency of 3.5 GHz as shown in fig 11, indicating that the surface provides a strong reflection response at this frequency. This value signifies efficient impedance matching and minimal power loss, ensuring that most of the incident electromagnetic energy is reflected rather than transmitted.

The transmission coefficient (S_{21}) was observed to be approximately –0 dB, confirming that the designed FSS effectively blocks transmission through the surface. This validates that the structure behaves predominantly as a reflective surface, which aligns with the project's focus on reflection-based operation for enhancing signal redirection in wireless environments.

Furthermore, the –10 dB bandwidth of the designed structure extends approximately from 3.36 GHz to 3.65 GHz, resulting in an effective bandwidth of around 283 MHz. This bandwidth range demonstrates that the FSS unit cell can operate efficiently across a significant portion of the 3.5 GHz 5G mid-band, ensuring stable performance under practical operating conditions.

The obtained simulation response confirms that the unit cell achieves the desired resonant characteristics with a well-defined passband and a sharp reflection profile.

Overall, the unit cell result establishes a strong foundation for further analysis, including angular stability and array formation. The excellent reflection performance and well-defined resonance confirm the suitability of the designed FSS for use in assisted 5G communication systems, where controlled reflection and phase manipulation are crucial for improving coverage and signal quality.

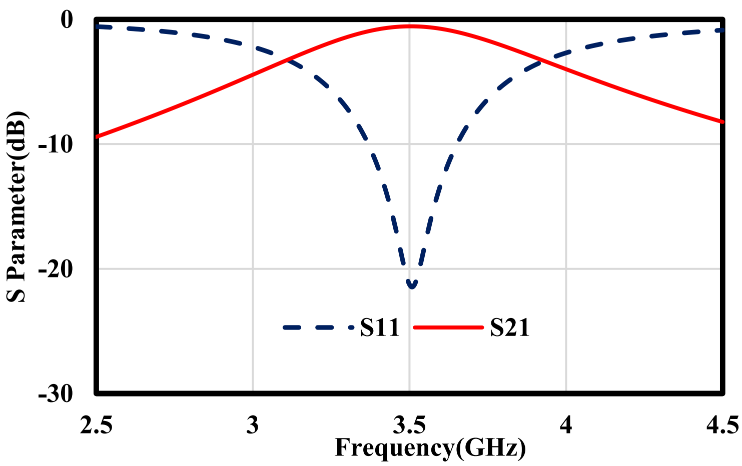


Fig.11. S-parameter resonance at 3.5 GHz

5.2 Angular Stability Analysis :

The angular stability of the designed Frequency Selective Surface (FSS) was analyzed for different phi (Φ) and theta (θ) angles ranging from 0° to 360° to ensure that the structure maintains stable reflection characteristics over varying incident angles as shown in fig 12 & 13.

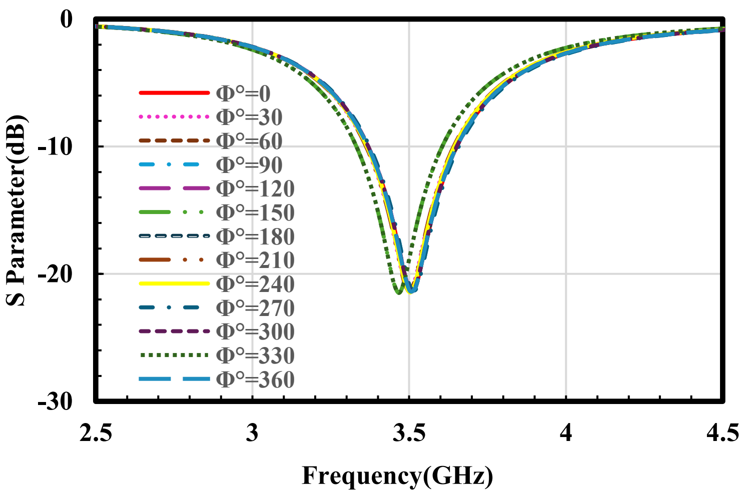


Fig.12. Reflection response for different phi angles

For Phi Variation:

The simulated reflection coefficient (S_{11}) for different φ (phi) angles ranging from 0° to 360° is shown in Fig. 12. It is observed that the resonant frequency remains nearly constant around 3.5 GHz for all φ values, indicating that the designed FSS unit cell exhibits good angular stability. The reflection magnitude shows only minimal variation throughout the entire φ range, confirming that the performance of the structure is stable and less sensitive to the change in azimuthal incidence.

This consistent behavior across all φ angles demonstrates that the FSS maintains its resonant characteristics irrespective of the rotation angle, which is a desirable property for reliable operation in practical wireless environments. The stable reflection response over 0°–360° ensures that the proposed structure can effectively reflect incident waves from different directions without significant frequency shifting or degradation in performance.

For Theta Variation:

The reflection response for different θ (theta) angles is illustrated in Fig. 13. It is observed that the resonant frequency remains stable around 3.5 GHz, with only slight variations in the reflection magnitude throughout the angular sweep from 0° to 360°. This indicates that the designed FSS structure exhibits excellent angular stability, maintaining its reflective characteristics even when the angle of incidence changes significantly.such stability across a wide range of θ angles confirms that the structure is capable of effectively handling oblique incidences without introducing frequency drift or degradation in reflection performance. This behavior is particularly advantageous in practical wireless communication scenarios, where electromagnetic waves may impinge on the surface from multiple directions due to reflections and scattering. The consistent reflection behavior over the entire angular range validates the robustness of the proposed design for use in Intelligent Reflecting Surface (IRS) applications, ensuring reliable operation under varying incident conditions.

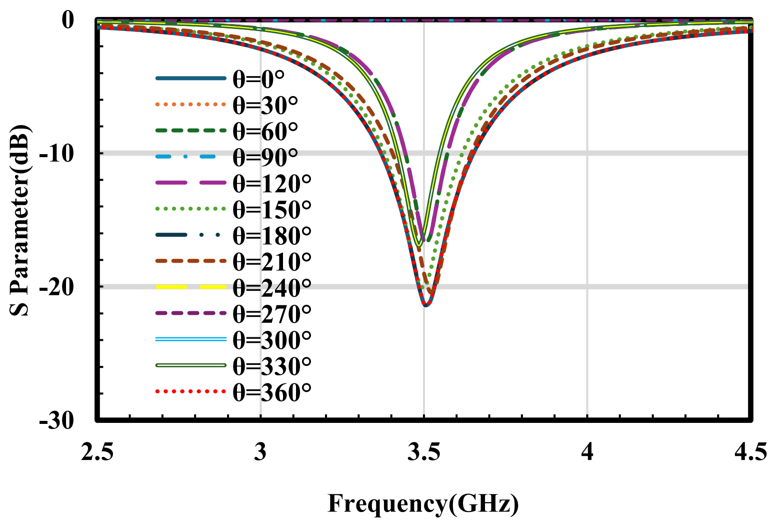


Fig.13. Reflection response for different theta angles

From both phi and theta analysis, it is evident that the designed FSS maintains its reflection characteristics across a wide angular range, making it suitable for 5G applications.

5.3 Array Structure Analysis :

To further validate the performance and scalability of the designed Frequency Selective Surface (FSS) unit cell, a 2 × 2 array configuration consisting of 4 identical unit cells was developed and simulated using HFSS with array size of 57.08 x 57.08. The main objective of forming this array was to analyze how the structure behaves when multiple elements interact electromagnetically and to confirm whether the reflection and resonance characteristics of the single unit cell are preserved in a larger, practical configuration suitable for real-world applications. In the array model, periodic boundary conditions were applied to ensure accurate field coupling between adjacent unit cells, replicating an infinite surface scenario. This configuration helps to study mutual coupling effects and observe any possible frequency shift or degradation in reflection performance due to array formation.

The simulated S-parameter response of the 2 × 2 array, shown in Fig. 14, demonstrates that the resonant frequency remains consistently centered at 3.5 GHz, which is in close agreement with the single unit cell response. The reflection coefficient (S_{11}) reaches a minimum of approximately –21 dB, indicating strong reflection and minimal signal loss at the desired frequency. This consistency between the single unit cell and array confirms that the electromagnetic behavior of the structure remains stable even when expanded to a multi-element configuration.

The transmission coefficient (S_{21}) is observed to be nearly 0 dB, confirming that almost all the incident electromagnetic energy is reflected and very little is transmitted through the surface.

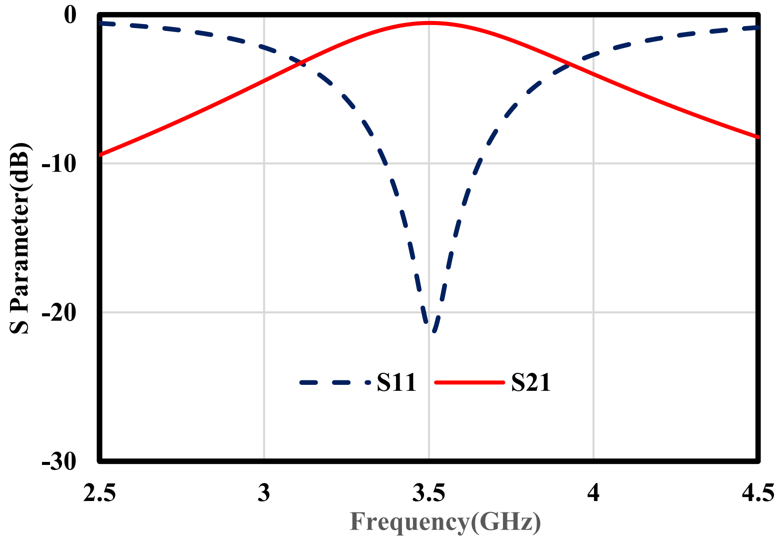


Fig.14. S Parameter of FSS array

Moreover, the array exhibits excellent reflection stability with negligible frequency drift or deformation in the resonant curve. This demonstrates that the designed unit cell is highly reliable when replicated across a larger surface, ensuring consistent performance over extended apertures. The array structure also improves the overall effective aperture area, enhancing the capability of the surface to interact with a broader incident wavefront, which is crucial in large-scale wireless coverage enhancement scenarios.

In conclusion, the array simulation results confirm that the proposed FSS design maintains its strong reflection characteristics, stable resonance, and minimal transmission loss when scaled up. This validates the suitability of the structure for integration where large arrays of such unit cells are employed to dynamically control reflection and phase distribution for 5G and beyond communication systems.

CHAPTER 6

FABRICTION PROCESS AND TESTING

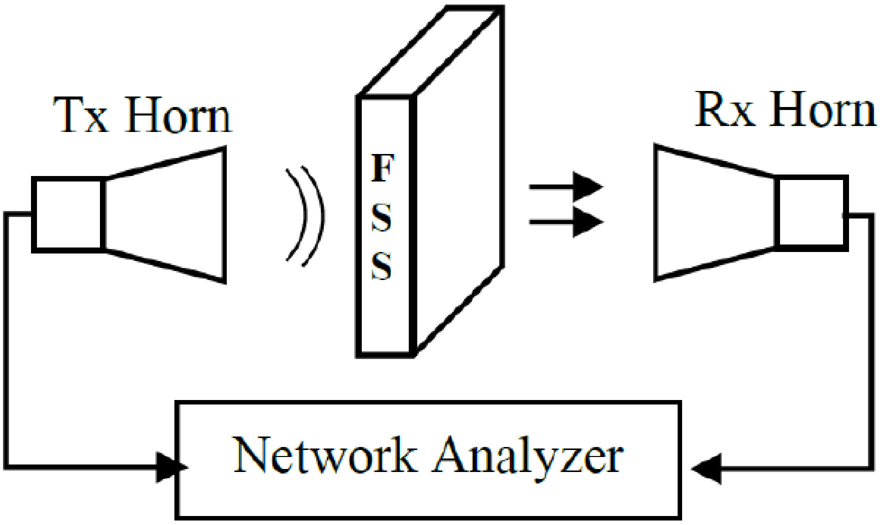


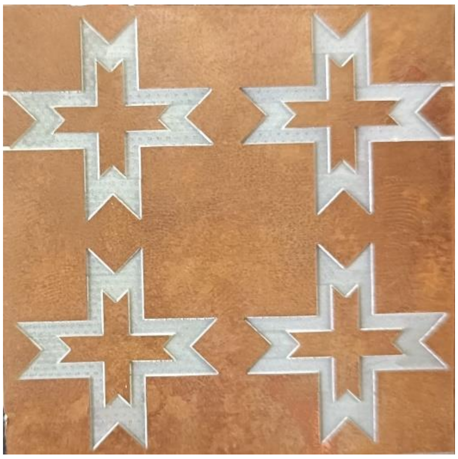
Fig.15. The experimental setup Block diagram

The proposed RF structure was successfully fabricated and tested using a controlled measurement setup. The fabricated sample was mounted on a rotating platform inside an RF anechoic chamber to ensure minimal external interference. The testing environment provides accurate measurement of electromagnetic characteristics by suppressing reflections and noise.

Testing Setup and Specifications

The measurements were carried out in an RF anechoic chamber equipped with a 3-axis automatic antenna measurement system. The chamber is lined with pyramidal RF absorbers to eliminate unwanted reflections and simulate free-space conditions.

The fabricated device under test (DUT) was placed on a rotating turntable to measure radiation characteristics at different angles. A standard horn antenna was used as the transmitting/receiving antenna. The system is connected to a Vector Network Analyzer (VNA) for S-parameter measurement.



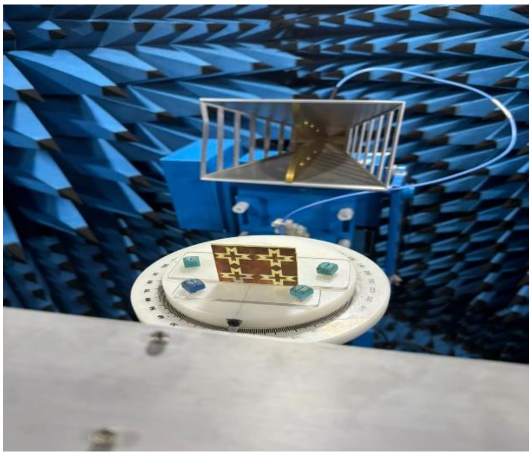


Fig.16. Fabricated FSS Prototype **Fig.17. FSS Under Measurement**

Specifications

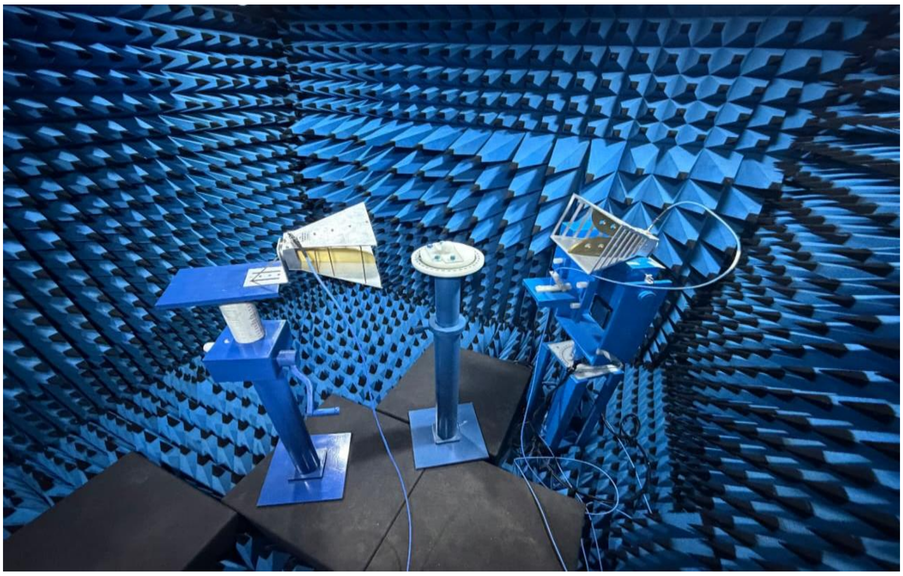


Fig.18. Anechoic Chamber setup

- Anechoic Chamber Dimensions: 3 m × 3 m × 2.6 m
- Frequency Range (Chamber): 800 MHz – 18 GHz
- Measurement System: 3-axis automatic antenna positioning system
- Instrument Used: Vector Network Analyzer (VNA)
- VNA Frequency Range: 1 MHz – 20 GHz

- Measurement Type: S-parameters (S11, S21) and radiation characteristics

Testing Result

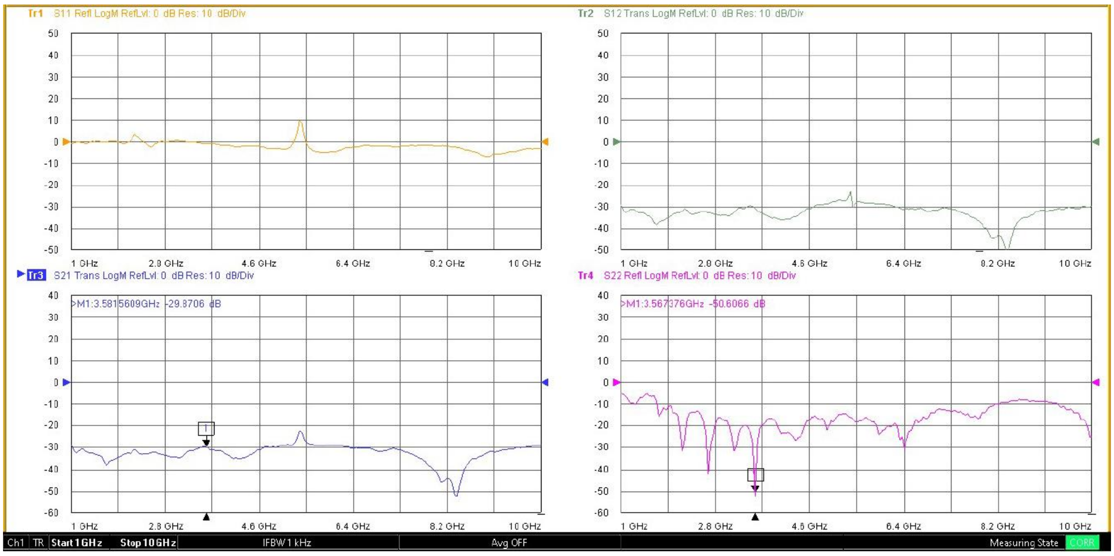


Fig.19. Fabricated FSS Testing Result

The fabricated Frequency Selective Surface (FSS) was experimentally characterized using a Vector Network Analyzer in an anechoic chamber environment. The measured S-parameters (S11, S21, S12, and S22) are presented in Fig. 19.

From the measurement results, it is observed that the proposed FSS exhibits a strong filtering response around 3.5 GHz. The transmission coefficient (S21) shows a value of approximately −29.87 dB at 3.58 GHz, indicating significant attenuation at the desired frequency band, thereby confirming the band-stop behavior of the structure.

Additionally, the reflection coefficient (S22) reaches approximately −50.60 dB at 3.56 GHz, demonstrating good impedance matching and minimal reflection loss at the operating frequency. The results also indicate stable performance across the measured frequency range.

The measured results are in good agreement with the simulated results obtained from Ansys HFSS, with slight variations due to fabrication tolerances, measurement setup, and environmental factors.

Overall, the fabricated FSS successfully validates the proposed design for 3.5 GHz sub-6 GHz applications, demonstrating effective filtering characteristics and practical feasibility.

S.No	ANGLE	Frequency Range (GHz)
1.	0°	3.56
2.	30°	3.56
3.	60°	3.56
4.	90°	3.56
5.	120°	3.56
6.	150°	3.56
7.	180°	3.56
8.	210°	3.56
9.	240°	3.56
10.	270°	3.56
11.	300°	3.56
12.	330°	3.56
13.	360°	3.56

Table 3 Different Angular Stability

The angular stability of the proposed Frequency Selective Surface (FSS) was analyzed by varying the azimuth angle (ϕ) from 0° to 360°. The corresponding resonant frequency values are presented in Table.

It is observed that the resonant frequency remains constant at 3.56 GHz for all angles. This indicates that the proposed FSS structure exhibits excellent rotational symmetry and polarization-independent behavior.

The consistent frequency response over the entire 0°–360° range confirms that the performance of the FSS is not affected by the orientation of the incident wave in the azimuth plane. Hence, the structure demonstrates high stability with respect to angular variation.

CHAPTER 7

CONCLUSION AND SCOPE OF PROJECT

7.1 Conclusion :

In this project, the design and analysis of a Frequency Selective Surface (FSS) operating at 3.5 GHz have been successfully carried out. The proposed FSS unit cell was designed and simulated using ANSYS HFSS to achieve the desired electromagnetic response. The structure was optimized to obtain effective performance in terms of reflection and transmission characteristics.

From the simulation results, the obtained S-parameters indicate that the designed structure exhibits the required frequency-selective behavior. The achieved S11 value of approximately –21 dB confirms good impedance matching at the target frequency, while the corresponding transmission characteristics validate the intended operation of the FSS. The bandwidth around the resonant frequency has also been analyzed, showing stable performance within the desired frequency range.

The study of current distribution (surface current magnitude and vector current) provided clear insight into how the structure interacts with electromagnetic waves, confirming that the geometry plays a crucial role in determining the resonant behavior. Additionally, the analysis of different incident angles (phi and theta) demonstrated that the structure maintains consistent performance under varying wave directions.

The design was further extended to an array configuration, which improves practical applicability and enhances real-world implementation. The fabricated structure and testing methodology using standard measurement techniques ensure that the proposed design can be validated experimentally.

Overall, the project successfully demonstrates the capability of FSS structures for selective frequency control, making them suitable for applications such as wireless communication systems, electromagnetic shielding, and urban signal management at sub-6 GHz frequencies.

7.2 Scope Of Project :

At The proposed Frequency Selective Surface (FSS) design at 3.5 GHz can be further extended in several directions to enhance its performance and applicability. Future work can focus on developing a reconfigurable FSS structure by incorporating active components such as PIN diodes or varactor diodes, enabling dynamic control of frequency response for adaptive communication systems.

The design can also be improved to support multi-band or wideband operation, making it suitable for multiple wireless standards including sub-6 GHz 5G and Wi-Fi applications. Additionally, the structure can be optimized for polarization independence and better angular stability to ensure consistent performance under different incident wave conditions.

Further research can explore the integration of the FSS with intelligent reflecting surfaces (IRS) to enhance signal coverage in complex environments such as urban areas. This can significantly improve signal strength, reduce interference, and support next-generation wireless communication systems.

On the fabrication side, advanced materials with low loss and flexible substrates can be used to improve efficiency and enable applications in wearable or conformal devices. The size of the structure can also be miniaturized using metamaterial concepts to make it more compact and efficient.

Finally, real-time experimental validation using advanced measurement setups can be carried out to compare simulation and practical results more accurately. This will help in refining the design and making it more reliable for commercial and industrial applications.

PROJECT OUTCOME MAPPED WITH SDG:

S.NO	PROJECT OUTCOME	SDG GOAL ACHIEVED(LOGO)
1	Enhances wireless communication efficiency by improving signal reflection, coverage, and link reliability in sub-6 GHz bands using IRS technology.	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
2	Supports the development of next-generation smart city networks by enabling better connectivity in dense urban areas with reduced dead zones.	11 SUSTAINABLE CITIES AND COMMUNITIES
3	Contributes to energy-efficient communication by reducing the need for excessive transmit power and minimizing network infrastructure load.	7 AFFORDABLE AND CLEAN ENERGY

4	Reduces electromagnetic wastage and improves spectral efficiency by intelligently controlling radio wave propagation.	

PROJECT OUTCOME MAPPED WITH POs & PSOs :

Project Title: Design Simulation and Testing of 3.5 GHz Band-Pass FSS for 5G Application

Domain: Radio Frequency

Programme Outcomes (POs) Mapping with Justification

PO No	Programme Outcome	Level (1–Low, 2–Medium, 3–High)	Justification (Specific to Project & Domain)
PO1	Engineering Knowledge	3	Applies core concepts of electromagnetics, RF engineering, and transmission line theory to design and analyze a Frequency Selective Surface (FSS) for desired frequency response..
PO2	Problem Analysis	3	Analyzes electromagnetic behavior and identifies challenges such as impedance mismatch, frequency selectivity, and unwanted signal attenuation to optimize FSS performance. mobility and safety; analyzes existing systems to develop an improved IoT-based solution.

PO3	Design/Development of Solutions	3	Designs and develops an FSS structure using unit cell configuration to achieve required band-stop or band-pass characteristics at target frequency.
PO4	Conduct Investigations of Complex Problems	2	Performs testing on sensor accuracy, environmental adaptability, and system response; analyzes results to enhance performance.
PO5	Modern Tool Usage	3	Utilizes modern RF simulation tools such as HFSS to design, simulate, and validate the FSS structure under different boundary and excitation conditions.
PO6	The Engineer and Society	3	Contributes to societal needs by enabling applications such as wireless communication, 5G filtering, and electromagnetic shielding using FSS structures.
PO7	Environment and Sustainability	2	Promotes efficient use of materials and energy through compact and passive RF structures, supporting sustainable communication system design.
PO8	Ethics	2	Ensures responsible engineering practices by maintaining accuracy in simulation results, proper reporting, and adherence to RF safety standards.
PO9	Individual and Team Work	2	Demonstrates effective teamwork in design, simulation, and analysis of the FSS structure, coordinating tasks across different stages of the project..
PO10	Communication	2	Communicates technical findings through structured reports, simulation results, and presentations explaining FSS design and performance.

PO11	Project Management and Finance	2	Applies planning and resource management in simulation, fabrication, and testing stages while maintaining cost-effective design.
PO12	Life-long Learning	2	Engages in continuous learning of advanced RF concepts, electromagnetic simulation tools, and emerging applications such as 5G.

Programme Specific Outcomes (PSOs) Mapping with Justification

PSO No	**Programme Outcome**	**Level (1–Low, 2–Medium, 3–High)**	**Justification (Specific to Project & Domain)**
PSO1	Analyze and Design Electronic Systems for Signal Processing and Communication Applications	3	Analyzes and designs a Frequency Selective Surface (FSS) structure to control electromagnetic wave propagation, achieving desired filtering characteristics for RF and communication applications.
PSO2	Apply Domain Specific Tools for Design, Analysis, Synthesis and Validation of VLSI and Communication Systems	3	Apply Domain Specific Tools for Design, Analysis, Synthesis and Validation of VLSI and Communication Systems (Level 2) Applies domain-specific tools such as HFSS for simulation, analysis, and validation of the FSS structure, ensuring accurate frequency response and performance optimization.

GAPC (Complex Engineering Problem) Table with WK Mapping

Sl. No	Complex Engineering Problem Attribute (P1–P7)	Description	Level (1–Low, 2–Medium, 3–High)	Mapped WKs	Justification (Project Specific)
1	P1: Depth of Knowledge Required	Involves use of advanced engineering knowledge	3	WK1, WK2	Requires strong understanding of electromagnetics, RF engineering, and wave propagation for designing FSS structures.
2	P2: Range of Conflicting Requirements	Trade-offs between multiple constraints	3	WK2, WK3	Balances frequency selectivity, bandwidth, insertion loss, and compact size while maintaining desired performance.
3	P3: Depth of Analysis Required	Requires analytical modeling and reasoning	3	WK2, WK4	Involves analysis of S-parameters (S11, S21) and electromagnetic response using simulation tools.
4	P4: Familiarity of Issues	Extent to which problem is new or familiar	2	WK2	The problem is partially familiar but requires customized FSS design for specific frequency applications like 5G.

Sl. No	Complex Engineering Problem Attribute (P1–P7)	Description	Level (1–Low, 2–Medium, 3–High)	Mapped WKs	Justification (Project Specific)
5	P5: Extent of Applicable Codes/Standards	Use of standards and regulations	2	WK6, WK8	Considers RF design guidelines, frequency standards, and safe electromagnetic operation.
6	P6: Stakeholder Involvement	Interaction with multiple stakeholders	3	WK6, WK7	Relevant to communication industries and users requiring efficient wireless systems.
7	P7: Interdependence of Subsystems	Integration of multiple subsystems	3	WK3, WK5	Involves integration of unit cell design, array configuration, and simulation environment for overall system performance.

WK (Knowledge Profile) Coverage Summary

WK No	Knowledge Area	Contribution Level	Remarks
WK1	Engineering Fundamentals	High	Strong application of electromagnetic theory, transmission lines, and RF fundamentals in the design and analysis of the FSS structure.
WK2	Problem Analysis	High	Involves detailed analysis of frequency response, S-parameters,

			and optimization of design parameters to achieve desired filtering characteristics.
WK3	Design/Development	High	Focuses on systematic design of unit cell and array configuration to meet specific band-pass or band-stop requirements.
WK4	Investigation	Medium	Includes simulation-based investigation and performance evaluation under varying conditions to validate design effectiveness.
WK5	Modern Tool Usage	High	Extensive use of electromagnetic simulation tools such as HFSS/CST for modeling, analysis, and validation of the FSS structure.
WK6	Engineer & Society	High	Contributes to development of efficient wireless communication systems used in modern applications like 5G and RF filtering.
WK7	Environment & Sustainability	Medium	Encourages compact and passive RF designs that reduce energy consumption and support sustainable communication technologies.
WK8	Ethics	Medium	Ensures accurate simulation, responsible reporting of results, and adherence to electromagnetic safety and communication standards.

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